

# Image Segmentation

- Image Segmentation is the process of partitioning a digital image into multiple regions (set of pixels).
- The goal of segmentation is to simplify and change the representation of an image into something that is more meaningful and easier to analyze.
- Segmentation accuracy determines the success or failure of computerized analysis procedure.
- Some practical applications of image segmentation are as follows:-

- i) It is used in medical imaging.
- ii) It is used to locate object in satellite image.
- iii) It is used in Face Recognition.
- iv) It is used in Fingerprint Recognition
- v) It is used for Machine Vision.
- vi) It is also used for Iris Recognition

- Image Segmentation algorithm are generally based on two properties;

- i) Discontinuity
- ii) Similarity (Continuity)

## (a) Discontinuity :

- Discontinuity is based to partition an image based on abrupt change in intensity such as edge in image.
- There are three basic types of gray level discontinuities in an image i.e, points or spot, lines and edges.



## b) Similarity:

→ Similarity is based to partition an image into regions that are similar according to predefined criteria (such as thresholding, region growing, region merging and splitting).

## 7.1. Detection of Discontinuities:

→ The most common way to look for discontinuities is to run a mask through the image.

→ For 3x3 mask the response (R) of the mask at any given point in the image is given by

$$R = w_1 z_1 + w_2 z_2 + \dots + w_9 z_9$$

$$R = \sum_{i=1}^9 w_i z_i$$

|       |       |       |
|-------|-------|-------|
| $w_1$ | $w_2$ | $w_3$ |
| $w_4$ | $w_5$ | $w_6$ |
| $w_7$ | $w_8$ | $w_9$ |

### 7.1.1. Point Detection:

→ Point detection is the principle of detection of isolated points in an image.

→ We can say that a point has been detected at the location on which the mask is centered if  $|R| \geq T$ ,

where  $T$  = ~~negat~~ non-negative threshold

$R$  = response of the mask.

→ The point can be detected using the mask shown below:

|    |    |    |
|----|----|----|
| -1 | -1 | -1 |
| -1 | 8  | -1 |
| -1 | -1 | -1 |

Fig. Point detection mask



→ This formulation measures the weighted difference between the centre point and its neighbors.

### 7.1.2. Line Detection:

→ Line detection is possible via various masks.

→ The horizontal mask,  $+45^\circ$  mask, Vertical mask and  $-45^\circ$  mask are shown below.

|    |    |    |
|----|----|----|
| -1 | -1 | -1 |
| 2  | 2  | 2  |
| -1 | -1 | -1 |

Fig. Horizontal mask

|    |   |    |
|----|---|----|
| 2  | 2 | -1 |
| -1 | 2 | -1 |
| -1 | 2 | -1 |

Fig. Vertical mask

|    |    |    |
|----|----|----|
| -1 | -1 | 2  |
| -1 | 2  | -1 |
| 2  | -1 | -1 |

Fig.  $+45^\circ$  mask

|    |    |    |
|----|----|----|
| 2  | -1 | -1 |
| -1 | 2  | -1 |
| -1 | -1 | 2  |

Fig.  $-45^\circ$  mask

### 7.1.3. Edge Detection:

→ Edge detection is based on discontinuity measurement.

→ It shows the abrupt change in the intensity of gray levels.

→ Edge can be defined as a set of connected pixels that lies in the boundary between two regions.

→ Derivatives are used for measurement of edges or edge detection.

→ Gradient and Laplacian are used for first and second order derivatives.



## a) Gradient Operator:

→ Gradient operator is based operator which computes first-order derivatives of a digital image like, Sobel Operator, Prewitt Operator, Robert Operator.

### (1) Sobel Operator:

→ Sobel operator is a discrete differentiation operator.  
→ Sobel operator computes the gradient approximation of image intensity function for image edge detection.  
→ Sobel operator uses two  $3 \times 3$  kernels or masks to calculate the vertical and horizontal derivative approximations respectively.

$$M_x =$$

|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
| -2 | 0 | 2 |
| -1 | 0 | 1 |

Fig. Vertical mask

$$M_y =$$

|    |    |    |
|----|----|----|
| -1 | -2 | -1 |
| 0  | 0  | 0  |
| 1  | 2  | 1  |

Fig. Horizontal Mask

### (2) Prewitt Operator:

→ Prewitt operator is almost similar to the Sobel operator.  
→ Prewitt operator ~~det~~ also detects vertical and horizontal edges of an image.  
→ Prewitt operator is one of the best way to detect the orientation and magnitude of an image.  
→ It also uses two kernels or mask as shown:

$$M_x =$$

|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
| -1 | 0 | 1 |
| -1 | 0 | 1 |

$$M_y =$$

|    |    |    |
|----|----|----|
| -1 | -1 | -1 |
| 0  | 0  | 0  |
| 1  | 1  | 1  |

### (3) Robert Operator:

- Robert operator computes the sum of squares of the differences between diagonally adjacent pixels in an image through discrete differentiation.
- Then the gradient approximation is made.
- It uses two  $2 \times 2$  kernels or masks as shown:

$$M_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$M_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

### b) Laplacian Operator (second order Derivative):

- The Laplacian of two dimensional image is given by

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- The above equation can be expressed in discrete form

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)]$$

- Cons: Laplacian is very sensible to noise; double edges.
- Pros: Dark or light side of the edge; zero crossings are of better use.



## 7.2. Edge Linking and Boundary Detection:

- An edge detection algorithm enhances the edges. When implemented, there are normally breaks in lines. Due to this reason, there are generally followed by linking procedures to assemble edge pixels into meaningful edges.
- There are two approaches for edge linking.

### (1) Local Processing:

→ Local Processing is the simplest approach for linking pixels in a small neighborhood.

→ The algorithm steps are as follows:-

step 1: Compute  $\nabla f$ ,  $M(x, y)$ ,  $\alpha(x, y)$

$$\text{where, } \nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$$

$$M(x, y) = \text{Mag.}(\nabla f) = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

$$\alpha(x, y) = \tan^{-1} \left( \frac{\partial f / \partial y}{\partial f / \partial x} \right)$$

step 2: Form a binary image.

$$g(x, y) = \begin{cases} 1 & ; M(x, y) > T_m \text{ and } \alpha(x, y) = A \pm T_a \\ 0 & ; \text{otherwise} \end{cases}$$

step 3: Scan the rows of  $g$  and fill all gaps in each row that do not exceed a specified length  $K$ .

step 4: Detect the gap in any direction  $\theta$ , rotate  $g$  by this angle and apply the horizontal scanning procedure in step 3. Rotate the result by  $-\theta$ .

## (2) Global Processing via the Hough Transform:

- Hough Transform is designed to detect lines, using the parametric representation of a line.
- Hough Transform is one of the powerful global method for detecting edges.
- Hough Transform can be used to connect the disjointed edge points.
- Hough Transform is used to fit the points as plane curves such as lines, circles and parabolas.

### → Algorithm for Hough Transform:

- (1) Load the image
- (2) Find the edges of the image using any edge detector.
- (3) Quantize the parameter space  $P$ .
- (4) Repeat the following for all the pixels of the image.

If the pixel is an edge pixel, then

(a)  $c = (-x)m + y$  or calculate  $\rho$

(b)  $P(c, m) = P(c, m) + 1$  or increment position in  $P$ .

(5) Show the Hough Space

(6) Find the local maxima in the parameter space.



(7) Draw the line using the local maxima.

→ The major problem with this algorithm is that it does not work for vertical lines, as they have a slope of infinity.

→ Solution: Convert line into polar coordinates  $\rho = x \cos \theta + y \sin \theta$ , where  $\theta$  is the angle between the line and x-axis and  $\rho$  is the diameter.



### 7.3. Thresholding:

→ Thresholding is one of the image segmentation techniques that generates a binary image from a given grayscale image by separating it into two regions based on a threshold value.

→ Thresholding may be categorized as:

#### (a) Local Thresholding:

→ A Thresholding is said to be local, if the Thresholding 'T' depends upon the gray levels of an image and some local properties also.

#### (b) Global Thresholding:

→ A Thresholding is said to be Global, if the Thresholding 'T' depends upon only the gray levels of an image.

→ To obtain T using global thresholding as follows:-

step 1: select an initial estimate for T.

step 2: Segment the image using T to form two groups of pixels  $G_1$  and  $G_2$ ;  $G_1$  consists of all pixels with gray levels  $> T$  and  $G_2$  consists of all pixels with gray levels  $\leq T$ .

step 3: Compute the average gray level of pixels in  $G_1$  to give  $\mu_1$  and  $G_2$  to give  $\mu_2$ .

step 4: compute a new threshold value:

$$T = \frac{\mu_1 + \mu_2}{2}$$

step 5: Repeat steps 2-4 until the difference in T in successive iterations is less than a predefined limit.



### (c) Adaptive Thresholding:

→ A Thresholding is said to be adaptive thresholding when the thresholding "T" depends upon the gray levels of an image, local properties and spatial coordinates of pixels. This is called dynamic thresholding.

→ Procedure for Adaptive Thresholding:

step 1: Convolve the image with suitable statistical operator i.e. mean or median.

step 2: Subtract the original from the convolved image.

step 3: Threshold the difference image with c.

step 4: Invert the threshold image.



## 7.4. Region Based Segmentation:

- Region-based segmentation groups pixels or regions based on their similar properties such as intensity, color or texture.
- This approach assumes that pixels within the same region have similar characteristics.
- This techniques are further classified into 2 types based on the approaches they follow:
  - (1) Region growing method
  - (2) Region splitting and merging method

### 7.4.1. Region Extraction:

- Suppose the spatial domain on which the image is defined is denoted by  $V$ .
- The image Segmentation techniques divides  $V$  into say  $n$  regions denoted by  $R_i$  ( $i = 1, 2, \dots, n$ ) such that
  - (1)  $\bigcup_{i=1}^n R_i = V$
  - (2)  $R_i \cap R_j = \emptyset$
  - (3)  $\text{Prop}(R_i) = \text{True}$
  - (4)  $\text{Prop}(R_i \cup R_j) = \text{False}$
- The first condition says that every pixel of the image domain is mapped to one region or the other.
- The second condition ensures that a pixel is mapped to only one region.
- The third condition indicates that the regions are defined on some property.
- Finally, the fourth condition assures maximality of each

### 7.4.2. Region Growing:

- Region Growing is a technique that groups pixels or sub-regions into larger region based on some pre-defined criteria.
- Region growing starts with a seed point and expands the region by adding neighboring pixels that have similar properties.
- The process continues until no more pixels can be added.
- Region growing is simple and intuitive, producing connected regions, but it is sensitive to noise and requires careful selection of seed points.

### 7.4.3. Region Splitting and Merging:

- Region splitting and merging is a hierarchical method that involves dividing the image into smaller regions and then merging adjacent regions with similar properties.
- Initially, the entire image is considered as a single region.
- The region is then recursively split until the resulting regions are homogeneous.
- Adjacent regions with similar properties are then merged.
- This technique is effective in handling complex images with varying intensity levels.



## # Image Segmentation based on Similarity:

- Image segmentation is the process of partitioning an image into segments or objects.
- The similarity approach in image segmentation groups pixels or regions based on their similar properties.
- Some common techniques based on similarity approach are as follows:

### (i) Thresholding:

- In thresholding, pixels are partitioned based on their intensity values.
- It involves converting a grayscale image into a binary image by applying a threshold value.
- Three types of thresholding: Local, Global and Adaptive Thresholding.

### (ii) Edge-Based Segmentation:

- This segmentation focuses on identifying the boundaries between different regions in an image.
- This technique detects significant changes in intensity or color, which typically indicate the presence of edges.
- It consists of different operators: Prewitt operator, Sobel Operator, Canny Edge Detector, Laplacian of Gaussian (LoG).

### (iii) Region-Based Segmentation:

- Region-based segmentation groups pixels or regions based on their similar properties such as intensity, color or texture.
- This approach assumes that pixels within the same region have similar characteristics.

### (iv) Clustering-Based Segmentation:

- Clustering-based segmentation partitions the image into groups (clusters) of similar pixels.
- This approach leverages unsupervised learning techniques to classify pixels based on their features, such as color, intensity or texture.

### (v) Artificial Neural Network-Based Segmentation:

- ANN-based segmentation leverages the power of deep learning to achieve high-precision segmentation results.
- This approach involves training neural networks on labeled datasets to learn the complex patterns and features within the images.



## 8. Representations and Description

### 8.1. Chain Codes:

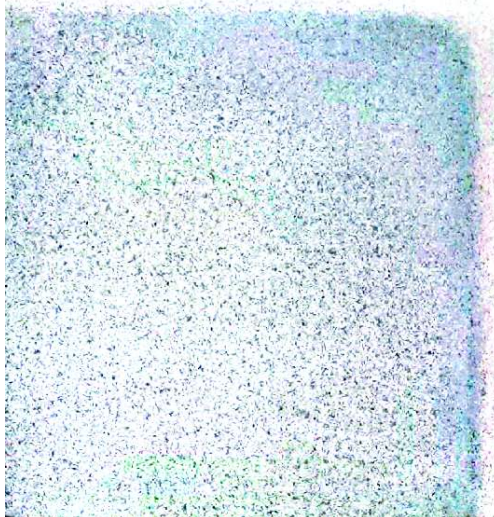
- Chain codes are the method for representing the boundaries of objects within binary images.
- They describe a boundary as a connected sequence of straight-line segments, each with a specified direction and length.

### 8.2. Signatures:

- A signature is a one-dimensional representation that captures the essential characteristics of a two-dimensional object's boundary or shape.

### 8.3. Shape Numbers:

- A shape number is a concise descriptor derived from the chain code of an object's boundary, capturing its geometric structure in a rotation-invariant manner.
- It is particularly useful for shape analysis and recognition tasks.





# Image Restoration.

4.

## 4.1. Introduction:

- Image Restoration is the class of methods that aims to remove or reduce the degradations that have occurred while the digital image was being obtained.
- The objective of image restoration is to improve a given image in some predefined sense.
- Image Enhancement is a subjective process whereas image restoration is an objective process.

## 4.2. A model of the Degradation/Restoration Process:

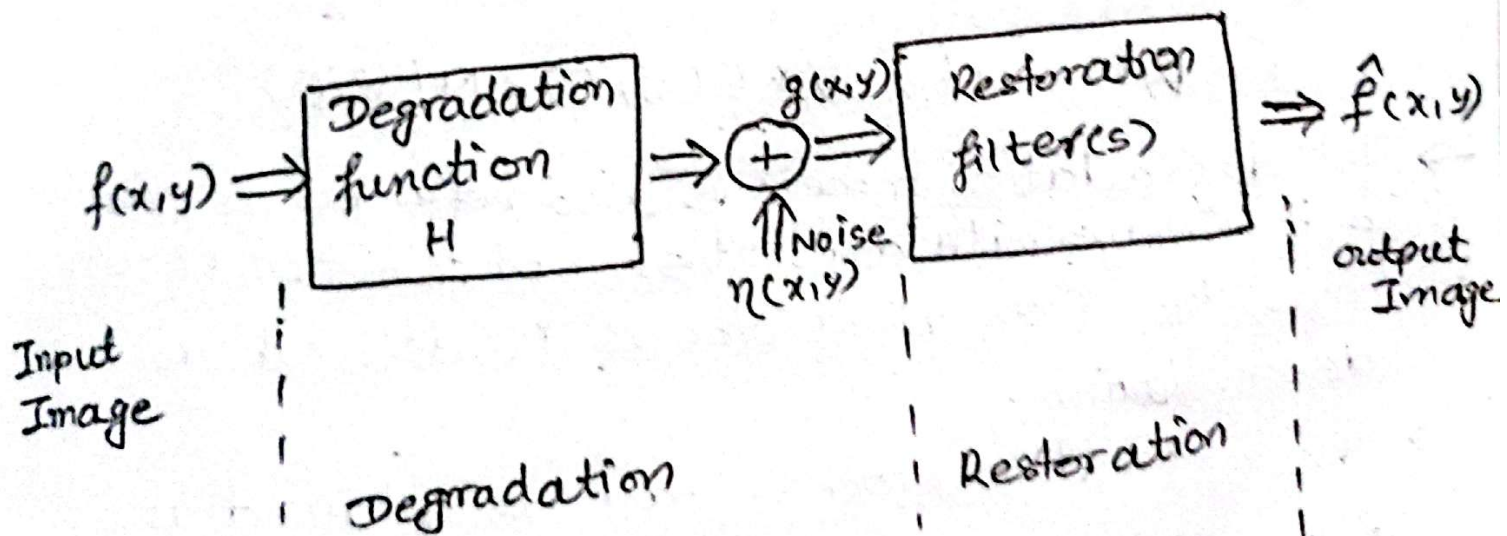


Fig. A model of the image Degradation/Restoration process



→ If  $f(x,y)$  and  $\eta(x,y)$  be the input image and additive noise terms respectively then, the degraded image  $g(x,y)$  is given by

$$g(x,y) = H[f(x,y)] + \eta(x,y)$$

→ The degraded image can be defined in spatial domain as,

$$g(x,y) = h(x,y) * f(x,y) + \eta(x,y)$$

where,  $h(x,y)$  = spatial representation of degradation function

→ The degraded image in spatial domain is equivalent to the convolution frequency domain and is represented as,

$$G(v,\theta) = H(v,\theta) * F(v,\theta) + N(v,\theta)$$

→ The degradation process is also referred as convolution of the image with a point spread function (PSF).

→ Thus, the restoration process is defined as deconvolution.

→ The restoration process can be obtained as,

$$\hat{f}(x,y) = R[g(x,y)], \text{ in linear spatial invariant form}$$

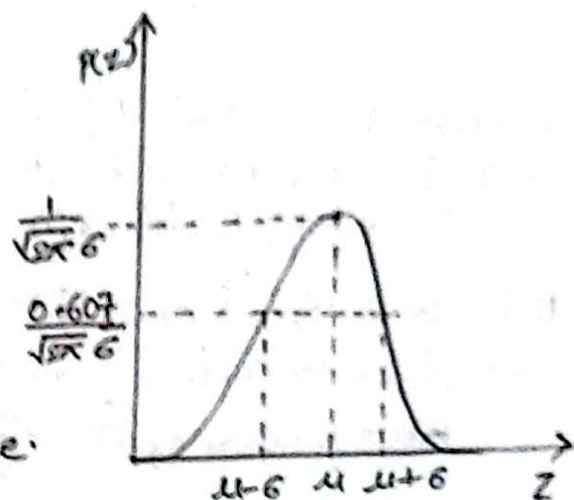
$$\hat{f}(v,\theta) = R[G(v,\theta)], \text{ in deconvolution form}$$

## 4.8. Noise Models in Image with PDF:

### i) Gaussian noise:

→ Gaussian noise models are simple to consider and the PDF of Gaussian random variable ( $z$ ) is given as,

$$P(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(z-\mu)^2/2\sigma^2}$$



#### a) Spatial Properties:

→ It appears as random variations with a smooth & continued distribution across image.

#### b) Frequency Properties:

→ It also affects both low and high frequency components.

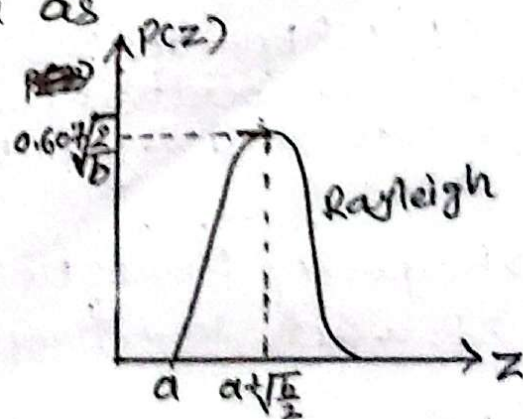
#### c) Filtering Technique:

→ Gaussian Smoothing or low pass filtering is effective for reducing Gaussian noise.

### ii) Rayleigh noise:

→ Unlike Gaussian distribution, the Rayleigh distribution is not symmetric. The PDF is given as

$$P(z) = \begin{cases} \frac{2}{b}(z-a)e^{-(z-a)^2/b} & ; z \geq a \\ 0 & ; z < a \end{cases}$$



#### \* Spatial Properties:

→ It manifests as amplitude variation in the image.

#### b) Frequency Properties:

→ It affects both low & high frequency components of an image.

#### c) Filtering Technique:

→ It uses Adaptive filters for reducing Rayleigh noise.



### iii) Gamma Noise (Erlang Noise) :

→ The PDF of Erlang (Gamma) Noise is given by

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & ; \text{for } z \geq 0 \\ 0 & ; \text{for } z < 0 \end{cases}$$

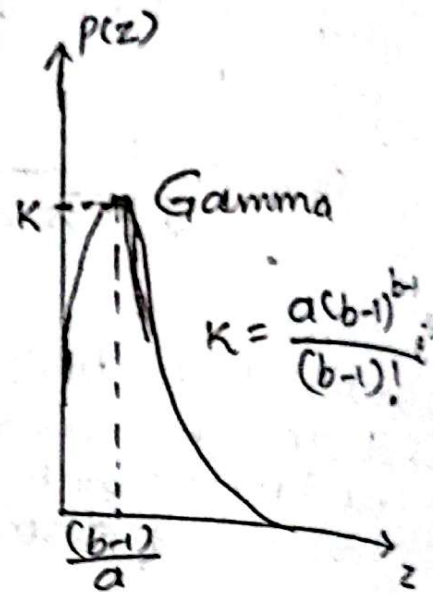
#### a) Spatial Properties:

→ It can manifest as variation in amplitude across image.

#### b) Frequency Properties:

→ It also affects both low pass & high pass frequency components.

#### c) Filtering Technique:



### iv) Exponential Noise:

→ The PDF of exponential noise is given as:

$$p_z(z) = \begin{cases} ae^{-az} & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$$

#### a) Spatial Properties:

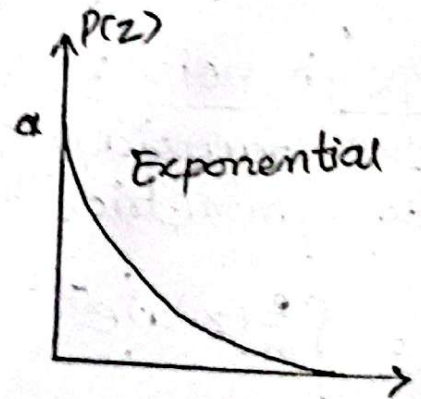
→ It is associated with random events occurring independently over time.

#### b) Frequency Properties:

→ It affects low frequency components of image or signal.

#### c) Filtering Technique:

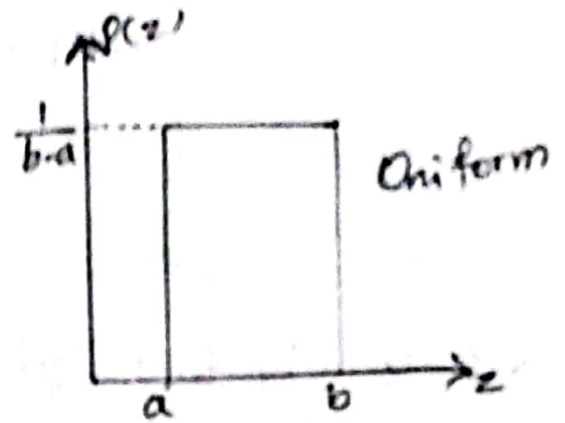
→ It uses low pass filtering for reducing noise.



### Uniform noise:

→ The PDF of uniform noise is given by

$$p(z) = \begin{cases} \frac{1}{b-a} & ; a \leq z \leq b \\ 0 & ; \text{otherwise} \end{cases}$$



#### a) Spatial Properties:

→ It manifests as constant value added to each pixel's intensity in image.

#### b) Frequency Properties:

→ It primarily affects low frequency components of an image.

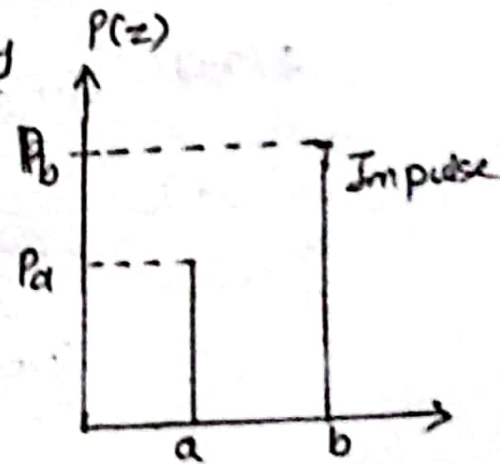
#### c) Filtering Technique:

→ It uses mean filtering technique for reducing noise.

### Salt and Pepper (Impulse) Noise:

→ The PDF of impulse noise is given by

$$p(z) = \begin{cases} p_a & ; \text{for } z=a \\ p_b & ; \text{for } z=b \\ 0 & ; \text{otherwise} \end{cases}$$



→ If  $b > a$ , gray level  $b$  will appear as a dot in image.

#### a) Spatial Filter Properties:

→ It appears as isolated, randomly occurring bright and dark pixels.

#### b) Frequency Properties:

→ It also affects both low & high frequency components.

#### c) Filtering Technique:

→



#### 4.4. Noise Models Restoration in the presence of Noise

##### Only Spatial Filtering:

→ The only degradation present in an image is noise i.e.

$$g(x,y) = f(x,y) + \eta(x,y)$$

$$\text{or, } G(u,v) = F(u,v) + N(u,v)$$

→ Following techniques can be used to reduce the noise effect:

i) Spatial Noise Filters

ii) Adaptive Spatial Filters

##### 4.4.1. Spatial Noise Filters:

→ In spatial noise filters, image independently varies from one location to another.

→ i) Arithmetic mean:

$$\hat{f}(x,y) = \frac{1}{mn} \sum_{(s,t) \in S_{xy}} g(s,t)$$

ii) Geometric mean:

$$\hat{f}(x,y) = \left[ \prod_{(s,t) \in S_{xy}} g(s,t) \right]^{1/mn}$$

iii) Harmonic mean:

$$\hat{f}(x,y) = \frac{mn}{\sum_{(s,t) \in S_{xy}} \frac{1}{g(s,t)}}$$

iv) Median:

$$\hat{f}(x,y) = \underset{(s,t) \in S_{xy}}{\text{median}} (g(s,t))$$

→ Max:

$$\hat{f}(x, y) = \max_{(s, t) \in S_{xy}} \{g(s, t)\}$$

→ Min:

$$\hat{f}(x, y) = \min_{(s, t) \in S_{xy}} \{g(s, t)\}$$

where  $m$  and  $n$  = variables denote image rows & columns spanned by the filter respectively.

$S_{xy}$  = denotes an  $m \times n$  sub image of the input noisy image  $g$ .

$\hat{f}(x, y)$  = filter response at  $(x, y)$  condition (coordinates)

$(x, y)$  = coordinates of sub image

#### 4.4.2. Adaptive Spatial Filters:

→ An adaptive spatial filter is one whose behavior can change based on statistical characteristics of an area within image.

→ An adaptive expression can be written as;

$$\hat{f}(x, y) = g(x, y) - \frac{\sigma_n^2}{\sigma_L^2} [g(x, y) - m_L]$$

→ The adaptive mean filtering algorithm uses two processing levels for adaptive spatial filters that can be denoted as;

is level A:

- If  $Z_{min} < Z_{med} < Z_{max}$  → go to level B

Else, increase the window size

- If  $Z_{window\ size} \leq S_{max}$ , repeat level A

Else output  $Z_{med}$ .



ii) Level B:

- If  $Z_{min} < Z_{xy} < Z_{max}$ , output  $Z_{xy}$   
Else output  $Z_{med}$

where

$Z_{min}$  = Min. Intensity value in  $S_{xy}$

$Z_{max}$  = Max. Intensity value in  $S_{xy}$

$Z_{med}$  = Median Intensity value in  $S_{xy}$

$Z_{xy}$  = Intensity value at co-ordinate  $(x, y)$

$S_{xy}$  = maximum allowed size of the adaptive filter window

→ It has three purposes:-

- i) Remove Impulse Noise
- ii) Provide Smoothing of other Noise
- iii) Reduce distortion

#### 4.5. Periodic Noise Reduction by Frequency Domain Filtering:

- Periodic noise is an image that arises typically from electrical or electromechanical interference during image acquisition.
- Periodic noise can be reduced typically significantly via frequency domain filtering.
- Periodic noise reduction technique uses a selective filter to isolate the noise.
- There are 3 types of selective filters used for periodic noise reduction:-

##### 1. Band Reject Filter:

- Band reject filter is used for removing periodic noise from an image involves removing a particular range of frequencies from that image.
- Types of Band reject filters:

##### a) Ideal Band Reject Filter:

- An ideal band reject filter is given as:-

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$

where

##### b) Butterworth Band Reject filter:

- A butterworth band reject filter is given as,

$$H(u, v) = \frac{1}{1 + \left[ \frac{D(u, v)W}{D^2(u, v) - D_0^2} \right]^{2n}}$$



## 1. Gaussian Band Reject Filter:

→ A Gaussian band reject filter is given by

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[ \frac{D^2(u, v) - D_0^2}{D(u, v) W} \right]^2}$$

## 2. Band Pass Filter:

→ A band pass filter performs the opposite operation of band reject filters.

→ A band pass filter allows a specific frequency band of the image to be passed and blocks the rest of frequencies.

→ The transfer function  $H_{BP}(u, v)$  of a band pass filter can be obtained from a corresponding band reject filter with transfer function  $H_{BR}(u, v)$  by using this equation,

$$\text{i.e. } H_{BP}(u, v) = 1 - H_{BR}(u, v)$$

→ These filters cannot be applied directly on an image because it may remove too much details of an image.

→ These are effective in isolating the effect of an image of selected frequency bands.

## 3. Notch Filters:

→ A notch filter rejects (or passes) frequencies in predefined neighborhoods about a center frequency.

→ Due to symmetry of the Fourier transform notch filters must appear in symmetric pairs about the origin.

→ The transfer function of an ideal notch filter of radius  $D_0$  with centers  $a(u_0, v_0)$  and by symmetry at  $(-u_0, v_0)$  is

$$D_1(u, v) = \sqrt{(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2}$$

$$D_2(u, v) = \sqrt{(u - M/2 + u_0)^2 + (v - N/2 + v_0)^2}$$

i) Ideal Notch Filter:

$$H(u, v) = \begin{cases} 0 & \text{if } D_1(u, v) \leq D_0 \text{ or } D_2(u, v) \leq D_0 \\ 1 & \text{otherwise} \end{cases}$$

$$D_1(u, v) = \sqrt{(u - M/2 - u_0)^2 + (v - N/2 - v_0)^2}$$

ii) Butterworth Notch Filter:

$$H(u, v) = \frac{1}{1 + \left[ \frac{D_0^2}{D_1(u, v) D_2(u, v)} \right]^{1/2}}$$

iii) Gaussian Notch Filter:

$$H(u, v) = 1 - e^{-\frac{1}{2} \left[ \frac{D_1(u, v) D_2(u, v)}{D_0^2} \right]}$$



Q. What are the main source of noise in image.  
Compare Image Enhancement and Image Restoration.

sg → The main source of noise in an image are as follows:-

- i) Sensor noise
- ii) Environmental noise
- iii) Transmission noise
- iv) Photographic noise

| S.N  | Image Enhancement                                                     | Image Restoration                                                             |
|------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------|
| i)   | It is a process which aims to improve bad images so they look better. | It is a process which aims to invert known or estimated degradation to image. |
| ii)  | It provides better visual representation.                             | It removes effects of sensing environment.                                    |
| iii) | It is a subjective process.                                           | It is an objective process.                                                   |
| iv)  | It is difficult to quantitative measures.                             | It is easy to quantitative measures by mathematical model.                    |
| v)   | It is concerned <del>with</del> about extraction of image features.   | It is concerned about restoration of degradation.                             |

## \* Algorithm for Haar Transform Kernel:

~~For N=2:~~

Step 1: Determine the order  $N$  of the Haar Basis.

Step 2: Determine  $n$  where  $n = \log_2 N$ .

Step 3: Determine  $p$  and  $q$ , where  $0 \leq p \leq n-1$

i) If  $p = 0$  then  $q = 0$  or  $q = 1$

ii) If  $p \neq 0$  then  $1 \leq q \leq 2^p$

Step 4: Determine  $K$ .

$$\text{i.e. } K = 2^p + q - 1$$

Step 5: Determine  $x$ .

$$x \rightarrow [0, 1] \rightarrow \left\{ \frac{0}{N}, \frac{1}{N}, \dots, \frac{N-1}{N} \right\}$$

Step 6:

$$h_0(x) = h_{0,0}(x) = \frac{1}{\sqrt{N}}, x \in [0, 1]$$

$$h_K(x) = h_{p,q}(x) = \frac{1}{\sqrt{N}} \begin{cases} 2^{p/2} & ; \frac{q-1}{2^p} \leq x \leq \frac{q-1/2}{2^p} \\ -2^{p/2} & ; \frac{q-1/2}{2^p} \leq x \leq \frac{q}{2^p} \\ 0 & ; \text{otherwise } x \in [0, 1] \end{cases}$$



The Haar Basis Matrix is given by

i) For  $N=2$ :

$$H_2 = \frac{1}{\sqrt{2}}$$

|   |    |
|---|----|
| 1 | 1  |
| 1 | -1 |

ii) For  $N=4$ :

$$H_4 = \frac{1}{\sqrt{4}}$$

|            |             |            |             |
|------------|-------------|------------|-------------|
| 1          | 1           | 1          | 1           |
| 1          | 1           | -1         | -1          |
| $\sqrt{2}$ | $-\sqrt{2}$ | 0          | 0           |
| 0          | 0           | $\sqrt{2}$ | $-\sqrt{2}$ |

*[Handwritten signature]*